Worksheet 1.2 — Geometric Sequences & Series

Mathematics: Analysis & Approaches SL (IB) · Unit 1: Number and Algebra

The Big Ideas

A geometric sequence has a constant common ratio r between consecutive terms.

- General term: $u_n = u_1 r^{n-1}$.
- Sum of the first n terms: $S_n = \dfrac{u_1(r^n - 1)}{r - 1}$, $r \neq 1$.
- Sum to infinity (only when $|r| < 1$): $S_\infty = \dfrac{u_1}{1 - r}$.

The Big Picture

When does S_∞ exist?

Only when $-1 < r < 1$ — otherwise the terms don't shrink toward 0 and the sum grows without bound.

Finding r from two terms

Divide two given terms to eliminate u_1: $\dfrac{u_m}{u_n} = r^{m-n}$, then solve for r.

Worked Examples

Example 1 — Find a term

Find u_9 for $3, 6, 12, \ldots$

Solution. $u_1 = 3$, $r = 2$. $u_9 = 3(2)^8 = 3(256)$.

Conclusion: $u_9 = 768$.

Example 2 — Find the common ratio from two terms

A geometric sequence has $u_1 = 500$ and $u_6 = 15.625$. Find r (assume $r > 0$).

Solution. $u_6 = u_1 r^5 \Rightarrow 15.625 = 500r^5 \Rightarrow r^5 = 0.03125 = \frac{1}{32}$.

Conclusion: $r = \frac{1}{2}$.

Example 3 — Sum to infinity

Find S_∞ for $40, -20, 10, -5, \ldots$

Solution. $u_1 = 40$, $r = -\frac{1}{2}$ (valid, since $|r| < 1$). $S_\infty = \dfrac{40}{1 - (-1/2)} = \dfrac{40}{3/2}$.

Conclusion: $S_\infty = \dfrac{80}{3}$.

Example 4 — Application — depreciation

A machine worth \$18,000 loses 20% of its value each year. Find its value after 5 years, and the total depreciation over that time.

Solution. **Step 1:** Value follows $u_n = 18000(0.8)^n$; after 5 years, $u_5 = 18000(0.8)^5 \approx 5898.24$.

Step 2: Total depreciation $= 18000 - 5898.24$.

Conclusion: value $\approx$ \$5898.24; depreciation $\approx$ \$12,101.76.

Example 5 — Solve for u_1 given S_infty

A geometric series has $r = 0.4$ and $S_\infty = 60$. Find u_1.

Solution. $60 = \dfrac{u_1}{1 - 0.4} = \dfrac{u_1}{0.6}$.

Conclusion: $u_1 = 36$.

Example 6 — Application — population growth

A town of 5000 grows by 3% each year. Find its population after 8 years (nearest whole number).

Solution. $u_9 = 5000(1.03)^8$ (year 8 growth applied 8 times from the start).

Conclusion: population $\approx$ 6333.

Example 7 — Sum of a finite geometric series with negative r

Find S_7 for $2, -6, 18, -54, \ldots$

Solution. $u_1 = 2$, $r = -3$. $S_7 = \dfrac{2((-3)^7 - 1)}{-3 - 1} = \dfrac{2(-2187 - 1)}{-4}$.

Conclusion: $S_7 = \dfrac{2(-2188)}{-4} = 1094$.

Example 8 — Solve for n given a term value

A geometric sequence has $u_1 = 5$, $r = 2$. Find n such that $u_n = 1280$.

Solution. $5(2)^{n-1} = 1280 \Rightarrow 2^{n-1} = 256 = 2^8$.

Conclusion: $n - 1 = 8 \Rightarrow n = 9$.

Example 9 — Application — doubling investment

An investment doubles every 6 years. If it starts at \$2000, express its value after t periods of 6 years as a geometric sequence, and find its value after 4 periods (24 years).

Solution. **Step 1:** $u_1 = 2000$, $r = 2$ per period. $u_5 = 2000(2)^4$ (value after 4 periods is the 5th term, counting the start as term 1).

Conclusion: $u_5 = 2000(16) =$ \$32,000.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find u_7 for $5, 10, 20, \ldots$

Question 2

A geometric sequence has $u_1 = 81$ and $u_5 = 1$. Find r (assume $r > 0$).

Question 3

Find S_{10} for $2, 6, 18, \ldots$

Question 4

Find S_∞ for $60, 12, 2.4, \ldots$

Question 5

Find S_∞ for $16, -8, 4, -2, \ldots$

Question 6

A ball dropped from 8 m rebounds to 60% of its previous height each bounce. Find the total vertical distance it travels (down and up), assuming it bounces forever.

Question 7

A savings account of \$2000 grows by 4% each year. Find its value after 10 years.

Question 8

A geometric series has $r = 0.25$ and $S_\infty = 80$. Find u_1.

Question 9

Find S_6 for $3, -9, 27, -81, \ldots$

Question 10

A geometric sequence has $u_1 = 4$, $r = 3$. Find n such that $u_n = 972$.

Question 11 — Challenge

A geometric sequence has $u_2 = 12$ and $u_5 = 324$. Find u_1 and r.

Question 12 — Challenge

Find S_∞ in terms of k for the sequence $k,\ \frac{k}{3},\ \frac{k}{9}, \ldots$ (assume $k > 0$), then find k if $S_\infty = 45$.

Question 13 — Challenge

A population of 8000 grows by 2.5% each year; find its population after 6 years (nearest whole number).

Question Recap & Answer Key

Question Recap

1. Find u_7 for $5, 10, 20, \ldots$
2. A geometric sequence has $u_1 = 81$ and $u_5 = 1$. Find r (assume $r > 0$).
3. Find S_{10} for $2, 6, 18, \ldots$
4. Find S_∞ for $60, 12, 2.4, \ldots$
5. Find S_∞ for $16, -8, 4, -2, \ldots$
6. A ball dropped from 8 m rebounds to 60% of its previous height each bounce. Find the total vertical distance it travels (down and up), assuming it bounces forever.
7. A savings account of \$2000 grows by 4% each year. Find its value after 10 years.
8. A geometric series has $r = 0.25$ and $S_\infty = 80$. Find u_1.
9. Find S_6 for $3, -9, 27, -81, \ldots$
10. A geometric sequence has $u_1 = 4$, $r = 3$. Find n such that $u_n = 972$.
11. A geometric sequence has $u_2 = 12$ and $u_5 = 324$. Find u_1 and r.
12. Find S_∞ in terms of k for the sequence $k, \frac{k}{3}, \frac{k}{9}, \ldots$ (assume $k > 0$), then find k if $S_\infty = 45$.
13. A population of 8000 grows by 2.5% each year; find its population after 6 years (nearest whole number).

✓ Answer Key

1. $u_7 = 320$
2. $r^4 = \frac{1}{81} \Rightarrow r = \frac{1}{3}$
3. $S_{10} = \frac{2(3^{10}-1)}{2} = 59048$
4. $S_\infty = \frac{60}{1-0.2} = 75$
5. $S_\infty = \frac{16}{1-(-1/2)} = \frac{32}{3}$
6. down 8, then up+down $2(8)(0.6) + 2(8)(0.6)^2 + \cdots = 2(8) \cdot \frac{0.6}{1-0.6} = 2(8)(1.5) = 24$; total $= 8 + 24 = 32$ m
7. $u_{10} = 2000(1.04)^9 \approx 2846.30$
8. $80 = \dfrac{u_1}{0.75} \Rightarrow u_1 = 60$
9. $S_6 = \frac{3(3^6-1)}{-4} = \frac{3(728)}{-4} = -546$
10. $4(3)^{n-1} = 972 \Rightarrow 3^{n-1} = 243 = 3^5 \Rightarrow n = 6$
11. $r^3 = \frac{324}{12} = 27 \Rightarrow r = 3$; $u_1 = 12/3 = 4$
12. $S_\infty = \dfrac{k}{1 - 1/3} = \dfrac{3k}{2}$; set $= 45 \Rightarrow k = 30$
13. $u_7 = 8000(1.025)^6 \approx 9277$